Assignment

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1. Caesar wants to arrange a secret meeting with Mark Antony, either at the Tiber River or at the Colosseum. He sends the ciphertext EVIRE. However, Antony does not know the key, so he tries all possible keys.

Given: Ciphertext = EVIRE

Caesar Cipher Encryption Formula: $C = (P + K) \bmod 26$

Decryption Formula: $P = (C - K) \bmod 26$

where K is the key and $0 \leq K \leq 25$

Cryptanalysis Model: Cipher-text only Attack

a. What are the meaningful plaintexts Antony obtains?

E = 4, V = 21, I = 8, R = 17, E = 4

Key, K = 1 : $(4-1) \bmod 26 = 3 = D$

$(21-1) \bmod 26 = 20 = U$

$(8-1) \bmod 26 = 7 = H$

$(17-1) \bmod 26 = 16 = Q$

$(4-1) \bmod 26 = 3 = D$

After trying all possible keys:

Key, K = 13 => Plaintext = RIVER

Key, K = 4 => Plaintext = ARENA

∴ the two meaningful plaintext have been found i.e., RIVER and ARENA

b. Where should Antony meet Caesar?

The two plaintexts indicate the possible meeting places:

- RIVER refers to the Tiber River.
- ARENA refers to the Colosseum, where gladiatorial contests took place.

Since Caesar's intended message is the Tibet River, Antony should meet Caesar at the Tibet River.

c. What interesting point about a ciphertext-only attack do you learn from this exercise?

- A ciphertext-only attack may produce more than one meaningful plaintext.
- The correct key may not be uniquely identified from a short ciphertext.
- Additional information, such as the context of the message, helps identify intended plaintext.
- Caesar cipher is weak because it has only 26 possible keys, making brute-force attacks easy.

2. Consider an Affine Cipher operating modulo 26. In a chosen-plaintext attack, the plaintext HAHAHA produces the ciphertext NONONO.

Plaintext = HAHAHA , Cryptanalysis Model : Chosen-Plaintext Attack

Ciphertext = NONONO

Encryption formula: $E(x) = (ax+b) \bmod 26$

a. Convert the plaintext and ciphertext letters into their numerical values.

Plaintext	H	A	H	A	H	A
Numerical Value	7	0	7	0	7	0

Ciphertext	N	O	N	O	N	O
Numerical value	13	14	13	14	13	14

b. Form the required equation modulo 26.

For H → N : $7a + b \equiv 13 \bmod 26$

For A → O : $0a + b \equiv 14 \bmod 26$

$b \equiv 14 \bmod 26$

c. Determine the values of a and b.

$7a + b \equiv 13 \bmod 26 \rightarrow ①$

$b \equiv 14 \bmod 26$

$b = 14$

Substitute value of b in eqn ①

$7a + 14 \equiv 13 \mod 26$

$7a \equiv 13 - 14 \mod 26$

$7a \equiv -1 \mod 26$

$7a \equiv 25 \mod 26$

$7^{-1} = 15$

$a \equiv (25 \times 15) \mod 26$

$a \equiv 375 \mod 26$

$a = 11$

$\therefore a = 11, \quad b = 14$

The key is valid, $\gcd(11, 26) = 1$

d. Write the encryption function $E(x) = (ax + b) \mod 26$.

$E(x) = (11x + 14) \mod 26$

For H $(x = 7)$,

$E(7) = (11 \times 7 + 14) \mod 26 = 91 \mod 26 = 13 = N$

For A $(x = 0)$,

$E(0) = (11 \times 0 + 14) \mod 26 = 14 = O$

$\therefore$ Hence valid.

3. The ciphertext CRWWZ was encrypted using an Affine cipher modulo 26. The attacker knows that the plaintext begins with HA.

Given: Ciphertext = CRWWZ

Plaintext begins with HA

We know, Encryption formula: $C = (aP + b) \bmod 26$

Cryptanalysis model : Known-plaintext attack.

a. Use the known plaintext - ciphertext pairs to determine a and b.

H = 7, A = 0

C = 2, R = 17

$\therefore E(7) = 2$

$E(0) = 17$

For H → C: $7a + b \equiv 2 \bmod 26$ — ①

For A → R: $0a + b \equiv 17 \bmod 26$

$b \equiv 17 \bmod 26$

$b = 17$

Substitute b = 17 in eqn ①

$7a + 17 \equiv 2 \bmod 26$

$7a \equiv -15 \bmod 26$

$7a \equiv 11 \bmod 26$

$7^{-1} \bmod 26 = 15$

$a \equiv 11 \times 15 \bmod 26$

$a \equiv 165 \bmod 26$

$a = 9$

$\therefore a = 9, \quad b = 17$

Key is valid, because $\gcd(9, 26) = 1$.

b. Write the encryption function.

$$E(x) = (9x + 17) \bmod 26$$

c. Find the multiplicative inverse of a modulo 26 and write the decryption function.

$9^{-1} \bmod 26$

$26 = 9(2) + 8$

$9 = 8(1) + 1$

$9 + 8(-1) = 1$

$9 + [26 + 9(-2)](-1) = 1$

$9 + 26(-1) + 9(2) = 1$

$9(3) + 26(-1) = 1$

$\therefore 3 \bmod 26 = \underline{3}$

$9^{-1} \bmod 26 = 3 \bmod 26 = 3$

Decryption Formula:

$$D(y) = a^{-1}(y-b) \mod 26$$

$$D(y) = 3(y-17) \mod 26$$

$$D(y) = (3y - 51) \mod 26$$

Since $-51 \equiv 1 \mod 26$

$$D(y) = (3y+1) \mod 26$$

d. Decrypt the complete ciphertext.

Ciphertext = CRWWZ.

$$P = (3C+1) \mod 26$$

$C \Rightarrow (3(2)+1) \mod 26 = 7 = H$

$R \Rightarrow (3(17)+1) \mod 26 = 52 \mod 26 = 0 = A$

$W \Rightarrow (3(22)+1) \mod 26 = 67 \mod 26 = 15 = P$

$W \Rightarrow (3(22)+1) \mod 26 = 67 \mod 26 = 15 = P$

$Z \Rightarrow (3(25)+1) \mod 26 = 76 \mod 26 = 24 = Y$

$\therefore$ Plaintext = HAPPY.

4. Alice uses a one-time pad to encrypt the plaintext RIVER with the key NQXTP.

Given : Plaintext = RIVER

Key = NQXTP

Cryptanalysis Model : No specific attack is required.

Encryption Formula => $C = (P + K) \bmod 26$

Decryption Formula => $P = (C - K) \bmod 26$

a. Convert the plaintext and key into numerical values.

Plaintext	:	R	I	V	E	R
Numerical Value	:	17	8	21	4	17

Key	:	N	Q	X	T	P
Numerical Value	:	13	16	23	19	15

b. Encrypt the message using $C_i = (P_i + K_i) \bmod 26$.

$C_1 = (P_1 + K_1) \bmod 26 = (17 + 13) \bmod 26 = 30 \bmod 26 = 4 = E$

$C_2 = (P_2 + K_2) \bmod 26 = (8 + 16) \bmod 26 = 24 \bmod 26 = 24 = Y$

$C_3 = (P_3 + K_3) \bmod 26 = (21 + 23) \bmod 26 = 44 \bmod 26 = 18 = S$

$C_4 = (P_4 + K_4) \bmod 26 = (4 + 19) \bmod 26 = 23 \bmod 26 = 23 = X$

$C_5 = (P_5 + K_5) \bmod 26 = (17 + 15) \bmod 26 = 32 \bmod 26 = 6 = G$

$\therefore$ Ciphertext = EYSXG.

C. Decrypt the resulting ciphertext using $P_i = (C_i - K_i) \bmod 26$.

$P_1 = (C_1 - K_1) \bmod 26 = (4 - 13) \bmod 26 = -9 \bmod 26 = 17 = R$

$P_2 = (C_2 - K_2) \bmod 26 = (24 - 16) \bmod 26 = 8 \bmod 26 = 8 = I$

$P_3 = (C_3 - K_3) \bmod 26 = (18 - 23) \bmod 26 = -5 \bmod 26 = 21 = V$

$P_4 = (C_4 - K_4) \bmod 26 = (23 - 19) \bmod 26 = 4 \bmod 26 = 4 = E$

$P_5 = (C_5 - K_5) \bmod 26 = (6 - 15) \bmod 26 = -9 \bmod 26 = 17 = R$

∴ Plaintext = RIVER.

d. Explain why the key must be random, equal in length to the plaintext and used only once.

- The key must be truly random so that each possible key letter is equally likely. A predictable key can make the plaintext easier to discover.
- The plaintext and key should have the same length, every plaintext character must have a corresponding key character. A shorter key would need to be repeated or reused, weakening security.
- If the same key encrypts two messages, relationship between the ciphertext can reveal information about their plaintexts. Reusing key compromises perfect secrecy.